



OPEN Effects of herbaceous root development on soil water infiltration and hydraulic properties in loess

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Root development alters soil pore structure and connectivity, thereby affecting un-saturated water movement; however, its growth-stage-dependent effects on infiltration and hydraulic properties in loess remain poorly quantified. In this study, one-dimensional soil column infiltration experiments were conducted on *Festuca arundinacea* root–soil composites under root distribution conditions corresponding to different growth stages ($T=0, 35, 52, 76$, and 99d). The results showed that total infiltration duration decreased significantly with increasing growth stage, indicating enhanced water transport capacity in loess. Saturated hydraulic conductivity showed a strong linear relationship with root biomass ratio (RB), while the other hydraulic parameters varied systematically with increasing RB and were well described by exponential functions. Based on the temporal evolution of root distribution, quantitative relationships between hydraulic parameters and growth stage were established and incorporated into the Richards equation. HYDRUS simulations agreed well with the measured wetting-front advance and volumetric water content changes. These findings indicate that herbaceous root growth stage is an important biological factor regulating infiltration behavior and hydraulic properties in loess, providing parameter support for studies of vegetation–soil–water interactions and soil water infiltration modeling under vegetation restoration on the Loess Plateau.

Keywords Growth stage, Root distribution, Root–soil composite, Soil water infiltration, Hydraulic properties

Water infiltration is a key process in unsaturated soil hydrology, directly controlling soil water redistribution, plant water uptake, rainfall runoff generation, and water cycling in slope ecosystems¹. In the Loess Plateau region of China, where precipitation is limited and highly uneven in both space and time, soil infiltration not only determines the efficiency of rainfall utilization but also profoundly affects vegetation restoration, soil conservation, and slope stability^{2,3}. Loess is characterized by a loose structure, uneven pore distribution, and pronounced softening upon wetting, and its infiltration behavior is therefore highly sensitive to changes in pore structure^{4,5}. Therefore, clarifying the biological and structural factors that affect infiltration in loess is of great importance for understanding soil hydrological processes and the effects of vegetation restoration in the Loess Plateau region.

Vegetation roots are important biological factors regulating soil pore structure and water movement. Previous studies have shown that roots can significantly affect water infiltration in unsaturated soils by altering pore connectivity, forming preferential flow pathways, and increasing the proportion of medium and large pores^{6–12}. Structural characteristics such as root diameter^{1,13–15}, distribution depth^{16–18}, and root content^{19–21} are closely related to improvements in soil pore structure and can further influence saturated hydraulic conductivity and the shape of the soil water characteristic curve (SWCC)^{10,22–24}.

However, most existing studies have focused on the effects of static root traits, such as root biomass, root-influenced depth, and root structure at a single time point, on soil hydraulic properties^{9,21,25}, whereas the temporal evolution of root architecture during plant growth and its hydrological effects remain relatively underexplored. In fact, root biomass, diameter, and distribution depth change continuously across growth stages, leading to stage-dependent variations in rhizosphere pore structure and soil hydraulic parameters^{19,26,27}. Most relevant studies have so far focused on woody vegetation through field observations or numerical analyses^{28,29}, whereas

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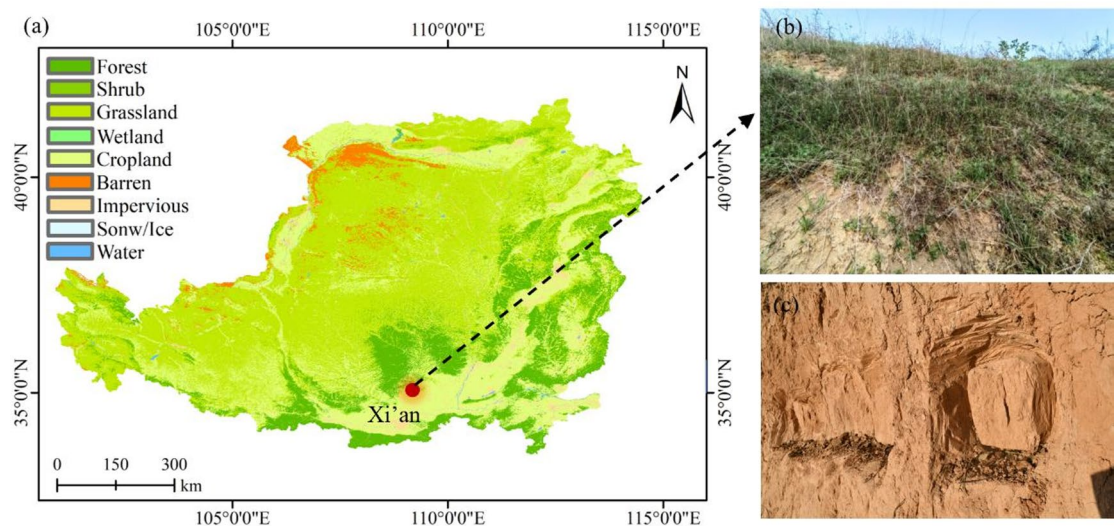


Fig. 1. Sampling position of loess sample. (a) Sampling point location diagram; (b) A loess vegetation slope in the study area; (c) Artificial sidewall soil.

Index	ρ_d (g/cm ³)	Natural water content (%)	Liquid limit (%)	Plastic limit (%)	Saturated water content (%)
Value	1.389	12.69	28.15	15.21	35.10

Table 1. Basic physical indexes of tested loess.

research on how root development in herbaceous plants affects water infiltration and hydraulic parameter dynamics in loess remains limited.

Under large-scale vegetation restoration, herbaceous roots have gradually become an important biological factor affecting pore structure and unsaturated water movement in loess^{25,30}. Therefore, systematically revealing the quantitative relationships between herbaceous root development and the unsaturated hydraulic parameters of loess, such as the Van Genuchten model parameters, and clarifying how root structural evolution across growth stages influences soil water infiltration and hydraulic properties are of great significance for a deeper understanding of vegetation–soil–water interactions and soil hydrological responses under ecological restoration in loess regions.

Based on the above, this study focused on *F. arundinacea* root–soil composites. Through root distribution investigation, one-dimensional soil column infiltration experiments, and unsaturated hydraulic parameter testing, the effects of herbaceous root development at different growth stages on soil water infiltration and hydraulic parameter evolution in loess were systematically investigated. Based on these results, quantitative relationships between hydraulic parameters and growth stage were established and incorporated into the Richards equation, and HYDRUS simulations were performed to evaluate the applicability and predictive capability of the proposed relationships for unsaturated infiltration under different growth stages.

Materials and methods

Basic properties of tested soil and root

The soil samples used in this study were collected from Bailuyuan, a representative area of the Loess Plateau in Xi'an, Shaanxi Province, China (Longitude: 109.096517°, Latitude: 34.214674°). This region is located in the transitional zone between the Guanzhong Basin and the southern edge of the Loess Plateau, with an annual average precipitation of approximately 500–700 mm. It is strongly influenced by a monsoon climate, with rainfall exhibiting marked seasonality, and short-duration, high-intensity rainfall events frequently occurring in summer^{33,34}.

As shown in Fig. 1(a), the vegetation in this area is dominated by herbaceous cover and cropland, exhibiting typical surface cover characteristics of the Loess Plateau. A natural slope in a public land was selected for sampling (Fig. 1(b)). After removing the surface loose soil, undisturbed loess samples were collected manually from the exposed sidewall of the slope (Fig. 1(c)). The basic physical properties of the loess were then determined, as listed in Table 1. The particle size distribution curve of the test soil is shown in Fig. 2. The tested soil was composed predominantly of silt, followed by clay, while sand accounted for the smallest proportion, showing a typical particle size composition of loess.

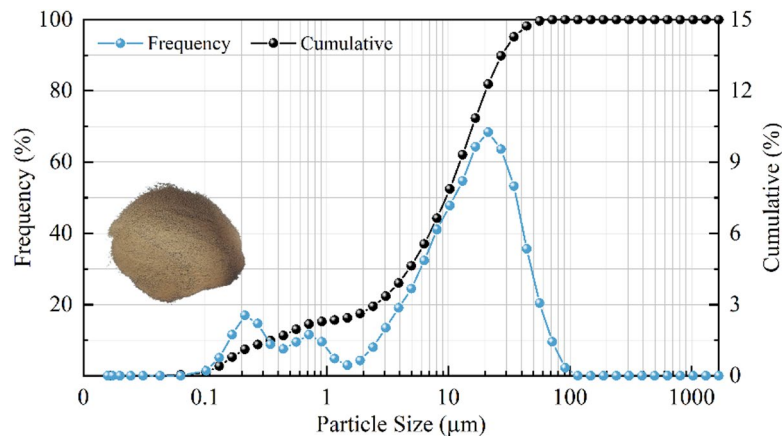


Fig. 2. Particle size distribution curve of the tested loess.

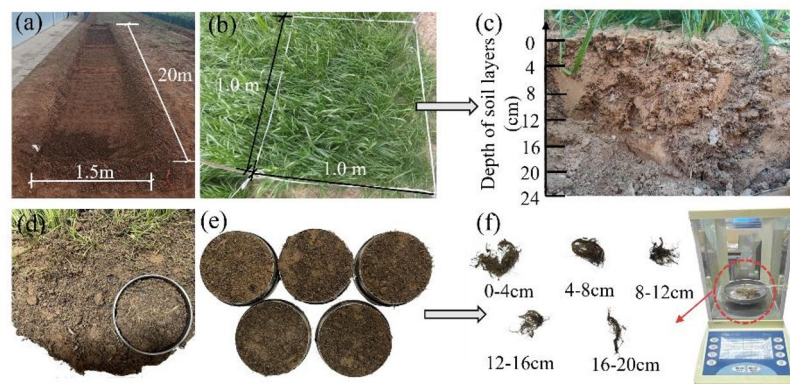


Fig. 3. Planting and sampling of *F. arundinacea*. (a): Planting area ($T = 0$ d); (b): Biomass surveys ($T = 52$ d); (c): Root profile of the planting; (d): Sampling; (e): Ring knife specimen; (f): Weighing the root samples at different strata.

Investigation of root distribution

Rainfall-induced shallow landslides in loess regions are predominantly distributed in areas covered by herbaceous vegetation, especially on the Loess Plateau, with previous studies reporting that approximately 69% of landslides occur in such areas^{31,32}. *F. arundinacea* is a commonly used herbaceous species for ecological slope protection on the Loess Plateau, is widely applied in soil–bioengineering practices such as highway, railway, and river embankment slope stabilization. Therefore, it was selected as a representative plant for this study to investigate the influence of herbaceous root development on soil water infiltration. To facilitate the collection of *F. arundinacea* roots at different growth stages and to observe their developmental patterns, a 30 m² rectangular planting area was established in this study (Fig. 3(a–b)). Roots of *F. arundinacea* at different growth stages were sampled, and their developmental characteristics are shown in Fig. 4. As shown in Fig. 4, during the 0–52 d stage, the root structure was relatively simple and had not yet differentiated into a complex architecture, while root length increased significantly with growth stage. When the growth stage reached $T = 64$ d, lateral root development became more pronounced, gradually forming a mature root network, and root length gradually stabilized at approximately 24 cm.

To further clarify the root distribution of *F. arundinacea* vegetation at different soil depths under different growth stages (T), layered sampling was conducted for *F. arundinacea* at different growth stages (0, 7, 15, 24, 35, 52, 64, 76, 88, and 99 d). First, the aboveground stems and leaves as well as the loose surface soil were removed. A ring knife ($\phi 6.18 \times 2$ cm) was then used to collect soil samples containing *F. arundinacea* roots from the 0–24 cm soil layer. The samples were grouped at 4 cm depth intervals, i.e., 0–4 cm, 4–8 cm, 8–12 cm, 12–16 cm, 16–20 cm, and 20–24 cm, resulting in six groups (Fig. 3(c)). During sampling, the cutting edge of the ring knife was kept facing downward and perpendicular to the ground surface (Fig. 3(d–e)) to ensure sampling reliability. The collected samples were then sieved to separate the roots, and the adhering soil on the root surface was removed by washing. The roots were air-dried to constant weight and then weighed (Fig. 3(f)). On this basis, the root biomass ratio (RB) of the root–soil composites at different soil depths was calculated by Eq. (1).

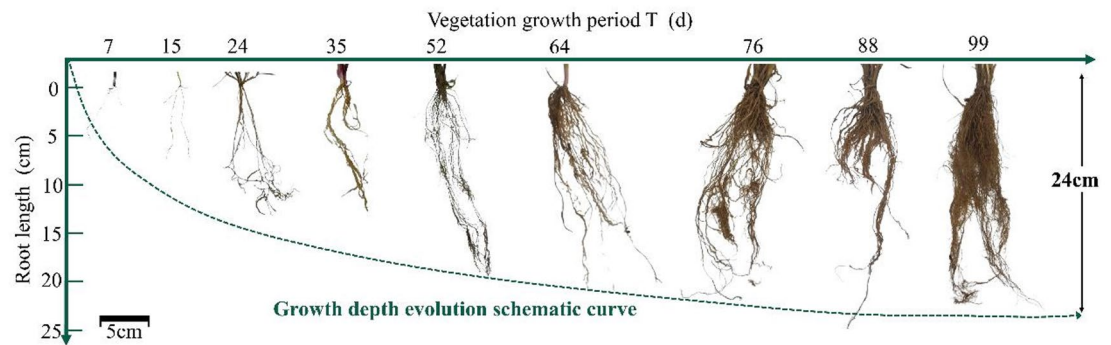


Fig. 4. Root distribution with depth at different growth stages of *F. arundinacea*.

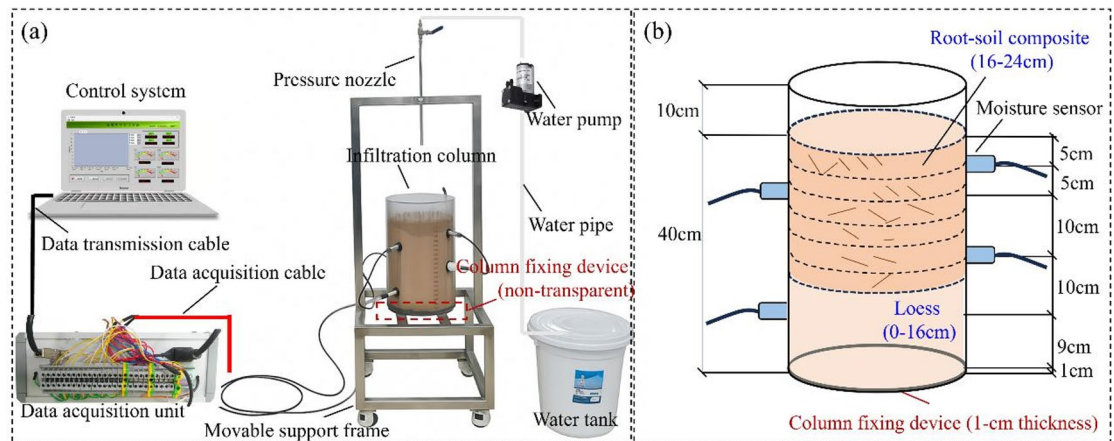


Fig. 5. Soil column infiltration test system. (a): Infiltration device physical map and (b): moisture sensor layout diagram.

$$RB = \frac{W_r}{V_s} \quad (1)$$

where, W_r is the total mass of roots contained in the ring-knife sample (g), and V_s is the volume of the ring-knife sample (mm^3).

Soil column infiltration experiment

In this study, a self-designed soil column infiltration apparatus (Fig. 5) was employed to conduct unsaturated water infiltration experiments on root–soil composites. The experimental setup mainly consisted of three components: an infiltration column system, a water supply system, and a data acquisition system, as illustrated in Fig. 5(a).

The infiltration column was a cylindrical container made of transparent acrylic, with a wall thickness of 1 cm, a height of 50 cm, and an inner diameter of 30 cm. A filter screen was installed at the bottom of the column to ensure uniform seepage distribution and to prevent the loss of fine particles. Four small openings were arranged on the sidewalls of the infiltration column (Fig. 5(b)), through which volumetric water content sensors were installed. The sensors used were CSF13 series soil water content sensors manufactured by XingYi (China), with a measurement range of 0–60% and an accuracy of $\pm 2\%$. During the experiments, the data sampling interval was set to 2 s.

Based on the root distribution characteristics of mature *F. arundinacea*, five growth stages ($T=0, 35, 52, 76$, and 99 d) were selected in this study to prepare root–soil composite column specimens. To realistically reproduce the actual root distribution pattern, a plain loess layer was arranged from the bottom of the column to a depth of 0–16 cm, while the root–soil composite layer was placed at depths of 16–40 cm (Fig. 5(b)). The root–soil composite section was further divided into six layers, with each layer being 4 cm thick. The RB assigned to each layer was determined according to the root distribution measured in the section “Investigation of root distribution” (the RB values for each layer are listed in Table S1).

During the preparation of the soil column specimens, the natural loess was air-dried, crushed, and sieved through a 2 mm mesh. Deionized water was then uniformly sprayed onto the soil using a spray bottle to achieve

the target water content. The moistened soil was subsequently sealed in plastic bags lined with plastic film and allowed to stand for 24 h to ensure uniform moisture distribution.

Based on the target dry density ρ_d (1.389 g/cm³), water content, cross-sectional area of the soil column, and the corresponding RB for each soil layer, the required masses of loose loess and *F. arundinacea* roots for each 2 cm-thick soil layer were calculated. The weighed loess and roots were thoroughly mixed to ensure homogeneity. The prepared loose soil mixture was then evenly placed into the infiltration column and compacted, resulting in a final layer thickness of 2 cm after compaction. After compaction, the surface of the soil layer was roughened to prevent layer separation during subsequent filling. This procedure was repeated layer by layer until a soil column specimen with a total height of 40 cm was formed.

After the preparation of the soil column specimens, the top of columns were sealed with plastic film and allowed to stand for 24 h. The initial volumetric water content of the soil column was then determined using the sensors (approximately 17.5% in this study), after which the infiltration experiments were conducted. Rainfall infiltration was applied using the water supply system, and to represent the short-duration, high-intensity summer rainfall commonly observed in the Bailuyuan area of the Loess Plateau, and based on previous studies^{21,35,36}, the rainfall intensity was set to 60 mm/h.

During the experiments, the wetting front migration was visually monitored through the transparent wall of the infiltration column. The experiment was terminated when the wetting front reached a position 1 cm above the bottom of the column, as the column fixing device at the base was non-transparent and obstructed further visual observation.

Infiltration parameter measurement

The water migration characteristics of unsaturated soils under rainfall can be described by the one-dimensional Richards equation, which incorporates the two dominant mechanisms governing water infiltration in unsaturated soils, namely gravitational potential and pressure potential, and takes soil water content as the only dependent variable³⁷. Previous studies have shown that the maximum water uptake rate of *F. arundinacea* roots is approximately $3.5\text{--}7.0 \times 10^{-5}$ cm³/cm³/min³⁸. During rainfall infiltration, the effect of root water uptake on water diffusion is very small and can be neglected. Therefore, the one-dimensional Richards equation for unsaturated root–soil composites can be expressed by Eq. (2)³⁷.

$$\frac{\partial \theta}{\partial t} + \frac{\partial}{\partial z} \left[K(\theta) - D(\theta) \frac{\partial \theta}{\partial z} \right] = 0 \quad (2)$$

where, t is the time parameter (s), z is the depth of soil layer (cm), $K(\theta)$ is the unsaturated hydraulic conductivity of the soil (cm/s), and $D(\theta)$ is the soil water diffusivity, which can be defined as $D(\theta) = K(\theta) \partial z / \partial t$.

When the Van Genuchten model is used to describe the SWCC of the root–soil composite, the unsaturated hydraulic conductivity $K(\theta)$ of the soil can be defined by Eq. (3)³⁵.

$$K(\theta) = K_s \cdot \frac{[1 - (\alpha S)^{n-1} [1 + (\alpha S)^n]^{-m}]^2}{(1 + (\alpha S)^n)^{1/2}} \quad (3)$$

where, K_s is the saturated hydraulic conductivity, θ_r is the residual water content, θ_s is the saturated water content, and S is the matric suction. The parameters α , m , and n are fitting parameters, with m defined as $m = 1 - 1/n$.

The incorporation of roots can have a significant impact on the unsaturated infiltration parameters of the soil (K_s , θ_r , θ_s , α , n). To clarify the evolution of these infiltration parameters with root content, five root–soil composite samples were prepared based on the measured root distribution patterns, corresponding to RB of 0%, 0.24%, 0.43%, 0.75%, and 0.96% (dry density $\rho_d = 1.389$ g/cm³). Matrix suction and saturated hydraulic conductivity tests were conducted on these samples. The preparation procedure was consistent with that used for the soil column experiments: trimmed *F. arundinacea* roots were thoroughly mixed with moist loess, and the root–soil mixture was compacted using molds. The resulting specimens were cylindrical, with a diameter of 6.18 cm and a height of 4.00 cm¹⁰.

To measure the matrix suction of the root–soil composites and construct the SWCC, ten water content levels were set for each RB treatment. During sample preparation, deionized water was added incrementally using a pipette to control the water content. After preparation, the samples were wrapped with cling film and left to equilibrate for 24 h to achieve uniform water distribution¹⁰. Matrix suction was determined using the axis translation technique, and saturated hydraulic conductivity was measured through laboratory constant-head tests.

Numerical calculation of soil column infiltration

Using the Hydrus-2D software³⁹, two dimensional soil column models were established under different growth stages T (0, 35, 52, 76 and 99 d), as shown in Fig. 6. The model domain was set to 30 cm $\times$ 40 cm, consistent with the laboratory experiments. The computational domain was divided into seven layers: the upper layer (0–24 cm) was assigned as the root–soil composite and further divided into six layers with a thickness of 4 cm each, denoted as Materials 1–6; while the lower layer (24–40 cm) was assigned as bare loess (Material 7). The corresponding material parameters were specified for each layer. The computational domain was discretized using an unstructured triangular finite element mesh, allowing effective representation of spatial heterogeneity while maintaining numerical stability. The final mesh consisted of 1456 nodes and 2798 two dimensional elements, with local refinement applied in key regions to ensure adequate resolution of hydraulic gradients

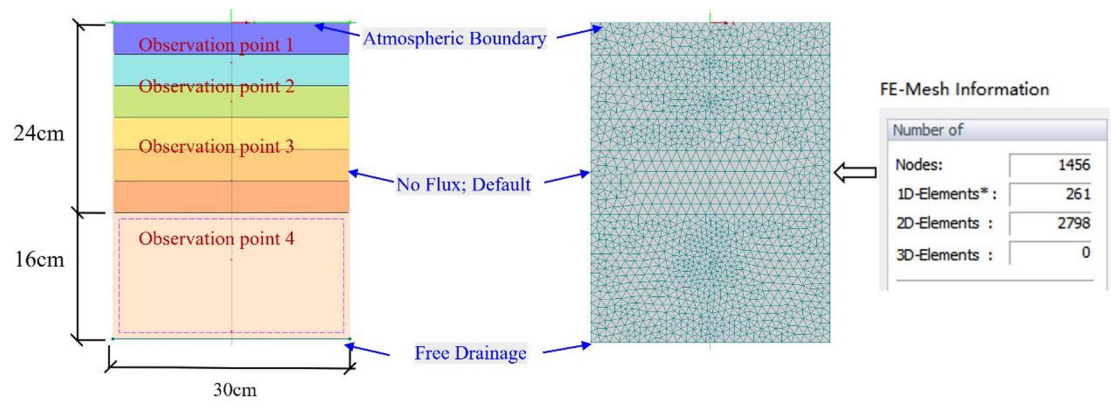


Fig. 6. Establishment of calculation model and setting of boundary conditions.

and infiltration processes. A free rainfall boundary condition was imposed at the top boundary, with a rainfall rate of 0.99 mL min^{-1} . The bottom boundary was specified as free drainage, while the lateral boundaries were defined as impermeable (Fig. 6). Observation points were placed at depths of 5, 10, 20, and 30 cm to monitor the temporal evolution of volumetric water content during infiltration. The infiltration process was terminated when the wetting front reached a depth of 39 cm.

Through numerical simulations, the wetting front and the temporal variation of volumetric water content of the root–soil composites under different growth stages can be obtained, which further allows evaluation of the reliability and applicability of the quantitative relationships of hydraulic parameters proposed in this study that account for the temporal evolution of root distribution.

Results

Evolution of root distribution

The distribution of root biomass ratio (RB) of *F. arundinacea* at different growth stages along the soil profile is shown in Fig. 7. As illustrated in Fig. 7, the root distribution range initially increases and then stabilizes with increasing growth stage. At $T = 7 \text{ d}$, *F. arundinacea* roots were only present in the 0–4 cm soil layer; at $T = 15 \text{ d}$, roots extended to the 0–8 cm layer; and at $T = 24 \text{ d}$, they reached the 0–12 cm layer. When the growth stage reached 52 d, the root distribution spanned 0–24 cm, and for stages beyond $T = 52 \text{ d}$, the distribution remained stable within 0–24 cm. The RB at each soil layer showed a general increasing trend with growth stage. Taking the 0–4 cm layer as an example, RB was minimal (0.0053%) at $T = 7 \text{ d}$, increased markedly to 0.6586% at $T = 64 \text{ d}$, and stabilized at approximately 0.8750% when T reached 88 d. During the preparation of soil column specimens for different growth stages, the RB value assigned to each soil layer was based on these measurements (the RB values for each layer are provided in Table S1).

The evolution of the RB of *F. arundinacea* roots within the 0–24 cm soil layer at different growth stages follows an exponential distribution, which can be described by Eq. (4).

$$\Psi_r = a_{RB} * \exp(-b_{RB} * \theta_h) \quad (4)$$

where, θ_h is the soil depth, and Ψ_r is equivalent to the previously defined RB and is denoted here as Ψ_r to standardize the parameters used in the theoretical derivation. Both a_{RB} and b_{RB} are experimental parameters related to vegetation growth stage, and their variations with growth stage are shown in Fig. 8.

As shown in Fig. 8(a), the experimental parameter a_{RB} first increases and then gradually approaches a stable value with increasing θ_T . Its evolution follows an S-shaped (Logistic) growth function^{42,43}, and can be described by Eq. (5).

$$a_{RB} = a_{RB}^{max} + \frac{a_{RB}^0 - a_{RB}^{max}}{1 + \left(\frac{\theta_T}{\theta_{TRB}}\right)^{P_{RB}}} \quad (5)$$

where, a_{RB}^{max} and a_{RB}^0 represent the maximum and minimum values of the experimental parameters during the growth period, taken as 0.013 mm and 0.0013 mm in this study, respectively. θ_T is a dimensionless time parameter, and θ_{TRB} is the midpoint of the increase rate, was taken as 46.5515 in this study. When $\theta_T < \theta_{TRB}$, the increase rate of root diameter rises with increasing θ_T ; and when $\theta_T > \theta_{TRB}$, the increase rate of root diameter decreases with increasing θ_T . P_{RB} is the growth rate parameter controlling the steepness of the curve, the larger the value of P_{RB} , the faster the curve increases, which were determined as 4.1951 in this study. As shown in Fig. 8(a), the calculated curve shows good agreement with the experimental results, with a coefficient of determination of $R^2 = 0.9961$.

As shown in Fig. 8(b), the experimental parameter b_{RB} exhibits an evolution trend of first decreasing rapidly and then gradually stabilizing with increasing θ_T , which follows an exponential function and can be described by Eq. (6).

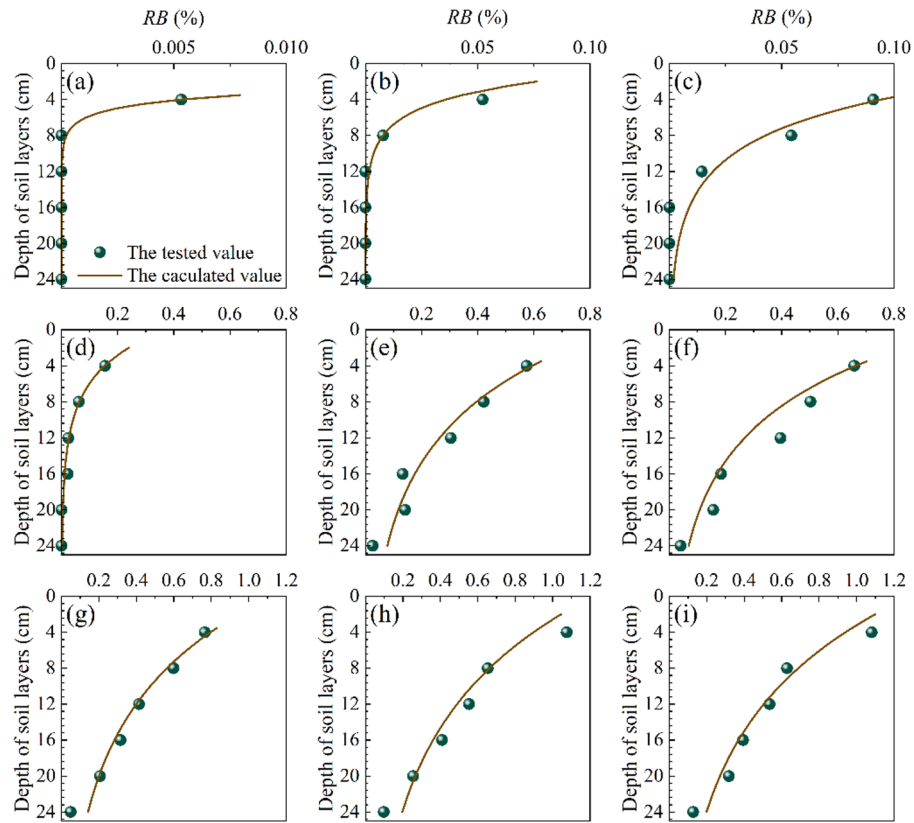


Fig. 7. Distribution map of root content in different growth stages. (a): $T = 7$ d; (b): $T = 15$ d; (c): $T = 24$ d; (d): $T = 35$ d; (e): $T = 52$ d; (f): $T = 64$ d; (g): $T = 76$ d; (h): $T = 88$ d; (i): $T = 99$ d.

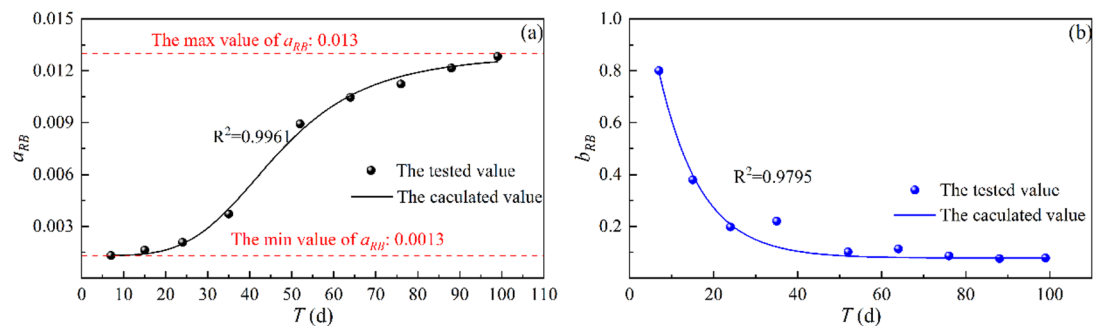


Fig. 8. Evolution law of test parameters. (a) a_{RB} ; (b): b_{RB} .

$$b_{RB} = b_{RB}^0 * X_{b1} * \exp\left(\frac{\theta_T}{X_{b2}}\right) \quad (6)$$

where, b_{RB}^0 is the minimum value of the experimental parameter b_{RB} during the growth period, which was taken as 0.00997 in this study. X_{b1} is the decay amplitude, representing the difference between the initial decay value and the offset, and was taken as 1.4425 in this study. X_{b2} is the decay rate constant, which controls the decay rate of the curve, and was taken as 9.5578 in this study. A comparison between the calculated curve and the experimental results is shown in Fig. 8(b), indicating a high degree of agreement, with a coefficient of determination of $R^2 = 0.9795$.

By combining Eqs. (4–6), a root distribution function considering vegetation growth stage can be obtained. This model can simultaneously account for the effects of soil depth and vegetation growth stage, and its calculated results agree well with the experimental results (Fig. 7).

Characteristics of wetting fronts and water variation

In the soil column infiltration experiments, water was applied from the top of the column. During infiltration, the interface between the wetted zone and the relatively dry zone is defined as the wetting front²¹, whose migration process could be clearly observed through the transparent wall of the column. Owing to obstruction by the fixing device, the observable range of the wetting front was limited to the zone extending from the top of the soil column to a depth of 0–39 cm.

Figure 9 presents that the wetting front migration of soil column specimens with different growth stages (T) during the infiltration process. The results show that, as infiltration proceeded, the wetting front gradually migrated downward. In addition, root incorporation effectively enhanced water movement in the soil, thereby altering the wetting-front migration pattern in the soil column specimens.

As shown in Fig. 9(a), for the bare loess specimen, the wetting front reached a depth of 2.52 cm at an infiltration time of $t = 10$ min and advanced to 13.50 cm at $t = 60$ min. The total infiltration duration was 403.8 min. With increasing T , the infiltration rate increased markedly. For the specimen with $T = 99$ d (Fig. 9(e)), the wetting front reached 17.75 cm at $t = 60$ min, which was 4.25 cm deeper than that of the bare loess. At $t = 120$ min, the wetting front advanced to 25.87 cm, exceeding that of the bare loess (18.28 cm) by 7.59 cm. Meanwhile, the total infiltration duration was reduced to approximately 272.93 min, representing a reduction of 47.95% compared with that of plain loess. Overall, as the growth stage increased, the total infiltration duration gradually decreased, and the wetting-front advance distance increased at the same infiltration time.

Figure 10 presents the variation in wetting-front advance rate with time under different growth-stage conditions. The wetting-front advance rate varied within a range of 0–0.5 cm/min. Under all growth stages T , the wetting-front advance rate showed a similar trend with increasing infiltration time: it first increased, reached a peak at $t = 10$ min, and then gradually decreased. To describe this evolution pattern, the wetting-front advance-rate curve was divided into three stages.

- (i) **Early infiltration stage:** During this stage, corresponding to $t = 0$ –10 min, the wetting-front advance rate increased rapidly and reached its maximum within 10 min. For plain loess, the peak infiltration rate was approximately 0.31 cm/min. As the growth stage increased, the RB in the soil also increased significantly, enhancing the promotion of water infiltration by roots and accelerating the wetting-front advance rate. At $T = 52$ d, the peak advance rate reached approximately 0.37 cm/min, representing an increase of 20.75% compared with plain loess. At $T = 76$ d, the peak advance rate reached approximately 0.43 cm/min, an increase of 38.16% relative to plain loess.
- (ii) **Mid-infiltration stage:** This stage corresponds to $t = 10$ –150 min, during which the wetting-front advance rate decreased rapidly. For the plain loess sample, the rate declined to 0.14 cm/min at $t = 150$ min, representing a reduction of approximately 54.77%. At the same time ($t = 150$ min), the wetting-front advance rates for soil columns at growth stages $T = 35, 52, 76$, and 99 d were 0.12, 0.14, 0.18, and 0.19 cm/min, respectively, all exceeding that of plain loess.
- (iii) **Late infiltration stage:** During this stage, the wetting-front advance rate decreased slowly and gradually approached a stable value. Taking the plain loess sample as an example, this stage lasted 253.8 min, and the advance rate gradually approached 0.10 cm/min. As the growth stage T increased, the RB within the 0–24 cm soil layer gradually increased, thereby enhancing the infiltration-promoting effect of roots. This was specifically reflected in a significant reduction in the total duration of stage (iii) and an increase in the stable advance rate with increasing T . For example, when $T = 52$ d, this stage lasted 200.4 min, and the advance rate was approximately 0.11 cm/min; when $T = 99$ d, the total duration decreased to approximately 122.93 min, while the advance rate increased to approximately 0.14 cm/min.

Characteristics of water variation within root–soil combinations

Figure 11 presents the responses of volumetric water content sensors installed at depths of 5, 10, 20, and 30 cm during the soil column infiltration experiments under different T . The results indicate that similar response patterns were observed at all monitoring depths. Specifically, when the wetting front reached a sensor location, the volumetric water content increased rapidly. After the wetting front passed the sensor, the rate of increase gradually decreased and eventually approached a stable value. The point at which a sudden increase in volumetric water content occurred corresponds to the sensor response time, denoted as S_I . Under the same T condition, S_I increased with increasing sensor depth. For example, at $T = 52$ d, the response times of the four sensors were 690 s, 1508 s, 5418 s, and 11,632 s, respectively.

Comparison of the S_I evolution curves under different T conditions (Fig. 11(f)) shows that, at all depths, the response time S_I decreased progressively with increasing T . And the response time S_I at each depth and the total infiltration duration gradually decreased with increasing T . When $T = 35$ d, the wetting-front advance rate was close to that of plain loess, and the sensor response times at all monitoring points also differed only slightly from those of plain loess. The total infiltration duration was approximately 23558 s, which was 670 s shorter than that of plain loess (24228 s), corresponding to a reduction of about 2.77%. This is because, at $T = 35$ d, *F. arundinacea* roots were sparse, distributed only within the 0–16 cm soil layer, and associated with relatively low RB values, resulting in a limited infiltration-promoting effect. As the growth stage T increased, the root distribution range of *F. arundinacea* gradually expanded, and RB increased rapidly at each soil layer, thereby significantly enhancing the infiltration-promoting effect of the roots. For example, when $T = 52$ d, the total infiltration duration was approximately 21,012 s, representing a reduction of 13.27% compared with plain loess; when $T = 76$ d, the total infiltration duration was approximately 17,568 s, a reduction of 27.49%; and when $T = 99$ d, the total infiltration duration was approximately 16,376 s, a reduction of 32.41%. The S_I values at all monitoring points also decreased significantly with increasing growth stage. Taking the monitoring result at a depth of 20 cm as an example, when $T = 35$ d, S_I was approximately 5814 s, which was 15.54% lower than that of

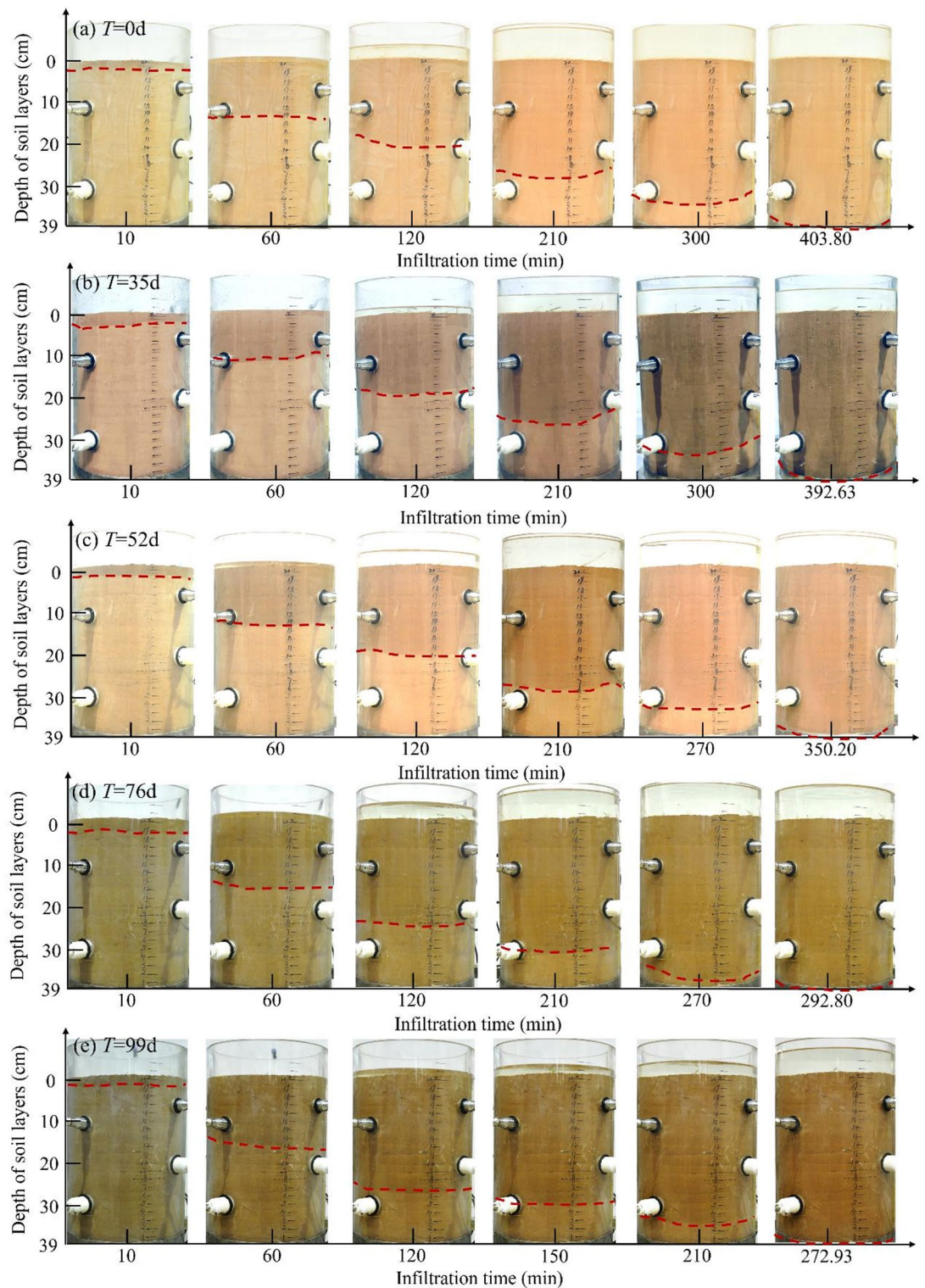


Fig. 9. Migration process of the wetting front in loess. (a): $T=0d$; (b): $T=35d$; (c): $T=52d$; (d): $T=76d$; (e): $T=99d$.

plain loess (6884s); when $T=52d$, S_I was 5418 s, representing a reduction of 33.81% relative to plain loess; and when $T=99d$, S_I at 20 cm reached its minimum value, approximately 4374 s, corresponding to a reduction of about 36.46% compared with plain loess.

Overall, mature roots can significantly improve soil structure and create more stable water flow pathways, thereby enhancing soil infiltration performance. Root incorporation can markedly improve the permeability of

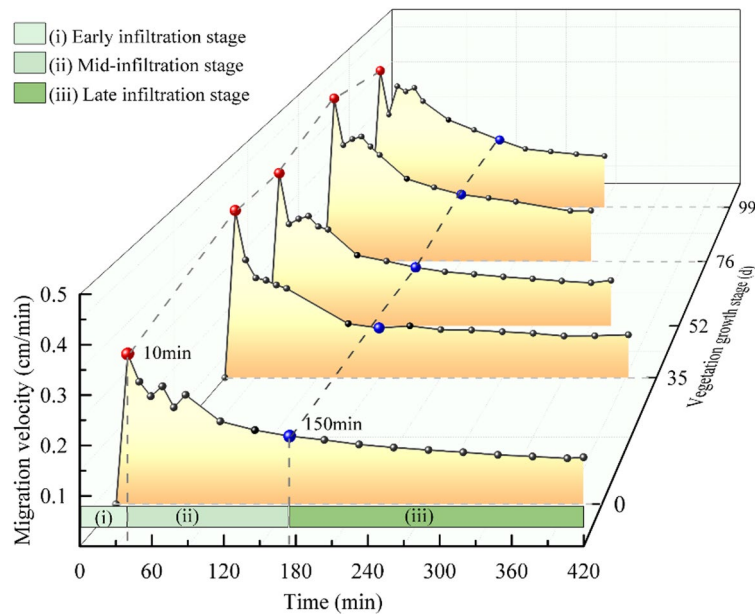


Fig. 10. The time-history curve of wetting front migration velocity with different T .

unsaturated loess, and the longer the growth stage and the higher the root content, the more pronounced the improvement in its infiltration properties.

Evolution law of infiltration parameters

Saturated hydraulic conductivity (K_s) of the root–soil composites under different RB was obtained through saturated permeability tests, as shown in Fig. 11(a). Matrix suction tests were conducted to determine the matrix suction distributions of plain loess and root–soil composites with four RB levels, as presented in Table S2 and Figure S1. By fitting the measured volumetric water content and matrix suction data to the Van Genuchten model⁴¹ using MATLAB, the evolution of saturated water content (θ_s), shape parameters (n) and (m), air-entry value (α), and residual water content (θ_r) of the root–soil composites under different RB conditions was obtained, as shown in Fig. 11(b–f).

Figure 12(a) shows that the saturated hydraulic conductivity (K_s) of the root–soil composites increased linearly with increasing RB . This is because root incorporation can effectively increase the proportion of medium and large pores in the soil, thereby providing lower flow resistance, higher connectivity, and more effective main flow pathways for water movement in saturated soil, which in turn significantly increases the saturated hydraulic conductivity K_s of the root–soil composites⁴⁰. The evolution of K_s with RB can be described by Eq. (7).

$$K_s = (a_{K_s} \cdot \Psi_r + 1) \cdot K_{s0} \quad (7)$$

where, K_{s0} is the saturated hydraulic conductivity of plain loess, which was approximately 7.606×10^{-5} cm/s in this study. a_{K_s} is an experimental parameter, which was taken as 1.4105. As shown by the comparison in Fig. 12(a), the calculated curve agrees well with the experimental results, with a coefficient of determination of $R^2 = 0.9622$.

As shown in Fig. 12(b–f), with increasing RB , the saturated water content θ_s of the root–soil composite exhibited an evolution trend of first increasing and then gradually stabilizing, whereas the shape parameters n and m , the air-entry value α , and the residual water content θ_r all showed trends of first decreasing and then gradually stabilizing. This is because θ_s , α , and θ_r are all closely related to the pore content of the soil. A higher pore content results in a larger θ_s ⁴⁴, but smaller values of α and θ_r ⁴⁵. After roots are incorporated into the soil, the pore characteristics can be effectively altered and the pore content can be increased^{32,46}, which in turn leads to changes in the above parameters. The evolution of θ_s , α , and θ_r with RB can be described by Eqs. (8–10).

$$\theta_s = \theta_{s0} + a_{\theta_s} \cdot \ln[b_{\theta_s} \cdot \Psi_r + 1] \cdot \theta_{s0} \quad (8)$$

$$\alpha = \alpha_0 - a_{\alpha} \cdot \ln[b_{\alpha} \cdot \Psi_r + 1] \cdot \alpha_0 \quad (9)$$

$$\theta_r = \theta_{r0} - a_{\theta_r} \cdot \ln[b_{\theta_r} \cdot \Psi_r + 1] \cdot \theta_{r0} \quad (10)$$

where, θ_{s0} , α_0 , and θ_{r0} are the saturated volumetric water content, air-entry value, and residual water content of plain loess, respectively, and were taken as 0.4602, 1.72, and 0.0162 in this study. a_{θ_s} , b_{θ_s} , a_{α} , b_{α} , a_{θ_r} , and b_{θ_r} are all experimental parameters, and were taken as 4.85×10^{-3} , 1.12×10^4 , 8.66×10^{-3} , 1.21×10^4 , 7.48×10^{-3} , and 1.47×10^5 , respectively, in this study. As shown by the comparison in Fig. 12(b–d), the

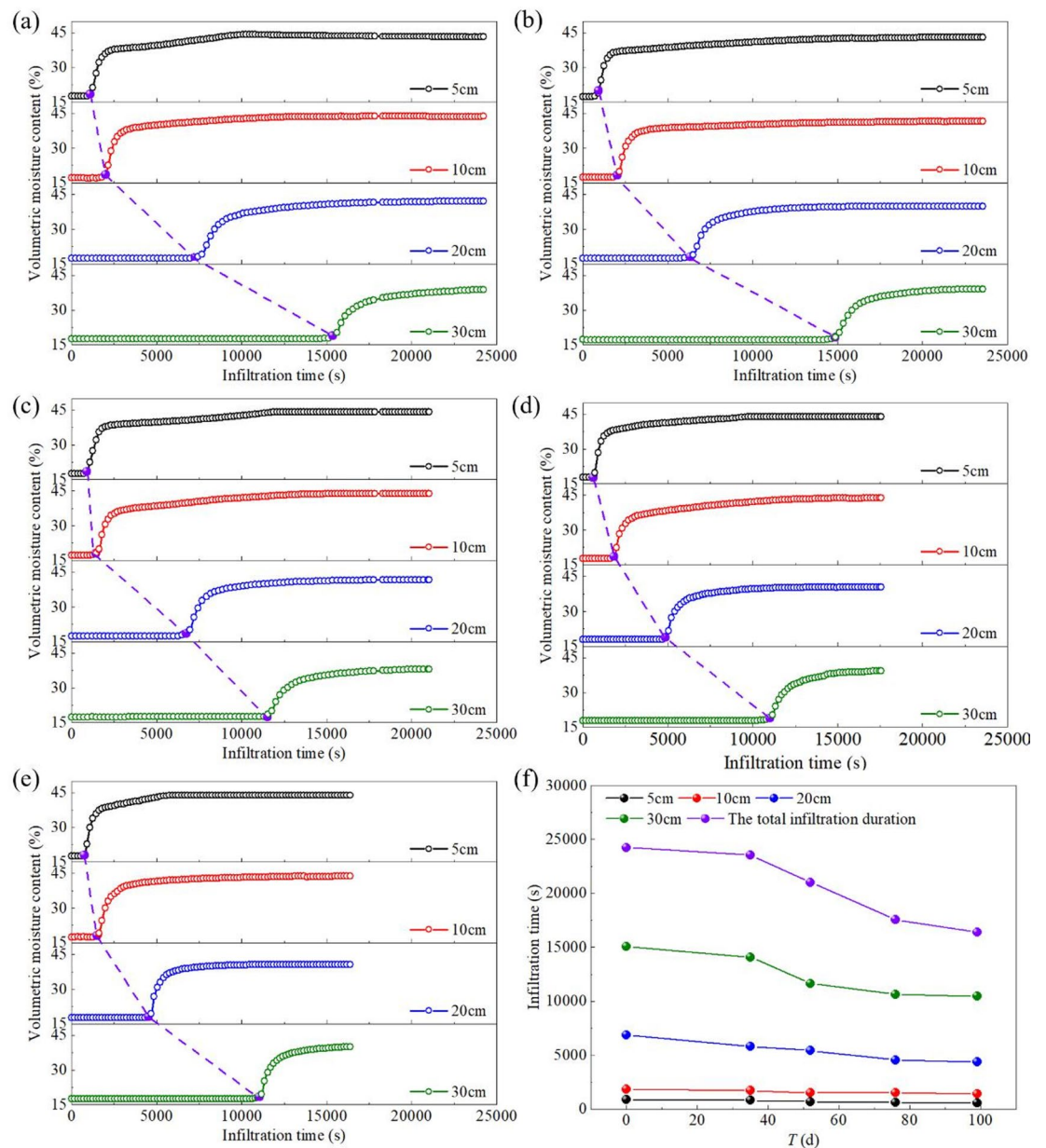


Fig. 11. Variation characteristics of volumetric water content in soil under different T . (a): $T = 0$ d; (b): $T = 35$ d; (c): $T = 52$ d; (d): $T = 76$ d; (e): $T = 99$ d; (f) S_I and infiltration time.

calculated curves agree well with the experimental results, with coefficients of determination R^2 of 0.9992, 0.9992, and 0.9990, respectively.

The shape parameters n and m control the form of the SWCC in the transition zone and are closely related to pore distribution characteristics. The more uniform the pore distribution, the larger the shape parameters become^{22–24}. Root incorporation can effectively alter the pore distribution pattern of the soil and reduce pore uniformity¹⁰, thereby causing changes in n and m . Their evolution with RB can be described by Eq. (11). The shape parameter m can be calculated as $m = 1 - \frac{1}{n}$.

$$n = n_0 - a_n \cdot \ln [b_n \cdot \Psi_r + 1] \cdot n_0 \quad (11)$$

where, a_n , b_n are experimental parameters, were taken as 5.38×10^{-3} , 1.19×10^5 in this study. As shown by the comparison in Fig. 12(e–f), the calculated curves agree well with the experimental results, with coefficients of determination R^2 of 0.9992 and 0.9990, respectively.

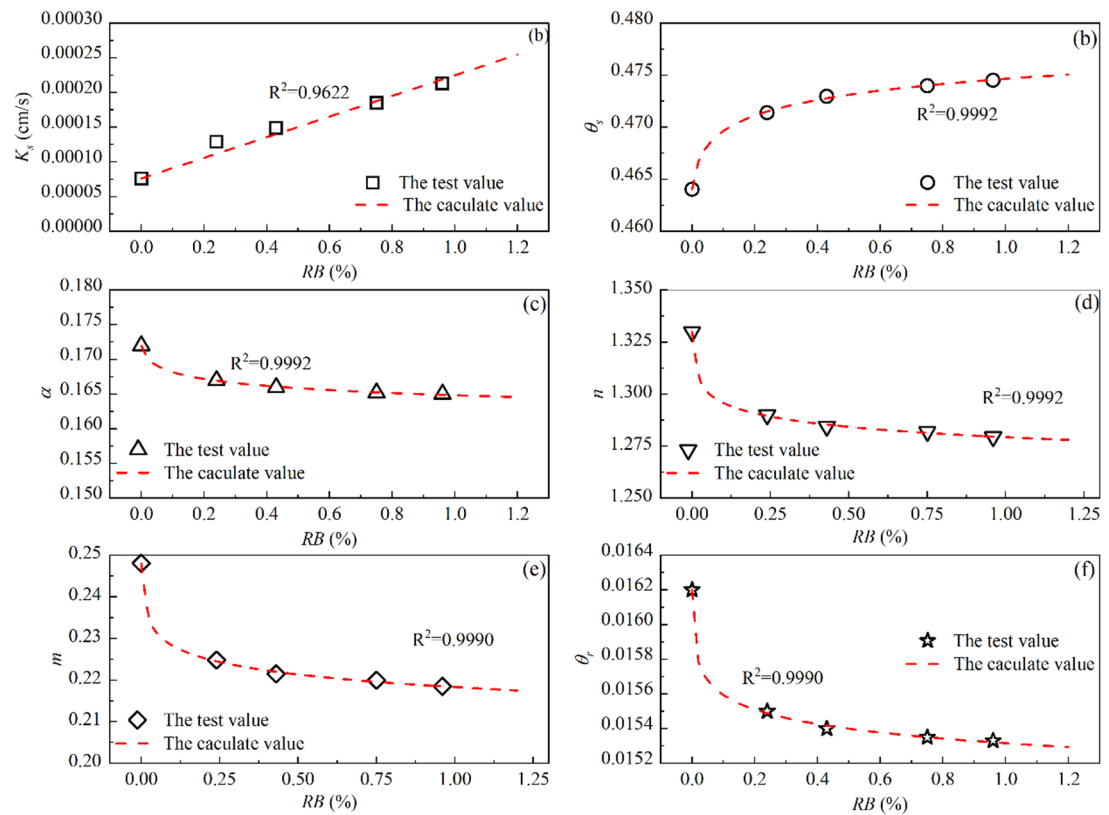


Fig. 12. Saturated permeability coefficient and calculated values of VG model parameters. (a): K_s , (b): θ_s , (c): α , (d): n , (e): m , (f): θ_r .

Discussion

Temporal response analysis of unsaturated infiltration parameters

Previous studies have demonstrated that root systems of various vegetation types significantly influence soil hydraulic properties. Trees and shrubs with deep and extensive root systems can enhance infiltration and saturated hydraulic conductivity in unsaturated soils through the formation of macropores and preferential flow paths^{34,35,47}. Other herbaceous plants primarily affect shallow soil layers, accelerating wetting front propagation and altering volumetric water content dynamics^{7,25}. These studies highlight that the magnitude, depth, and spatial distribution of root effects vary among vegetation types, which should be carefully considered when evaluating soil water transport processes. The *F. arundinacea* roots used in this study are characterized by small diameters, dense distribution, and highly flexible networks. When incorporated into the soil, these roots, through mechanisms such as network formation and compression, promote the formation of local soil aggregates, alter the pore structure³². Specifically, root incorporation effectively increases the total porosity and the proportion of medium and large pores. This increase in medium-to-large pores enhances pore connectivity, while decreasing the uniformity of pore distribution^{10,40}. Consequently, these changes modify the soil infiltration parameters, including K_s , saturated water content, and residual water content^{32,40}. The root distribution is closely related to its growth stage T . With increasing T , both the root-influenced depth and root content gradually increase. Based on Eqs. (7–11), the evolution of unsaturated infiltration parameters of the root–soil composite at different soil depths under the influence of *F. arundinacea* roots at different growth stages (0, 35, 52, 76, and 99 d) was calculated, as shown in Fig. 13.

Figure 13 presents the calculated results, during the growth stage of $T=0$ –52d, *F. arundinacea* roots affected only the soil layer within 0–12 cm, and their influence on the saturated hydraulic conductivity of the soil was limited (Fig. 13(a)). Taking $T=35$ d as an example, the infiltration-promoting effect of roots was confined to the 0–8 cm soil layer. At 0–4 cm, the saturated hydraulic conductivity was approximately 8.38×10^{-5} , which was only 10.22% higher than that of plain loess (8.38×10^{-5}); at 4–8 cm, it was 8.42×10^{-5} , representing an increase of only 1.53%. Meanwhile, parameters such as the air-entry value α and the shape parameters n and m also changed only slightly. Therefore, under the growth stages of 35 and 52 d, the soil column infiltration tests exhibited only a slight reduction in total infiltration duration compared with plain loess, and the sensor response times at different soil depths remained close to those of plain loess.

When the growth stage exceeded 52 d, the influence range of *F. arundinacea* roots rapidly extended to the 0–24 cm soil layer, and both the infiltration-promoting range and the infiltration-promoting effect increased significantly. For example, at $T=76$ d, the infiltration parameters had changed markedly. Taking the saturated hydraulic conductivity K_s as an example, K_s increased significantly throughout the 0–24 cm soil layer. At

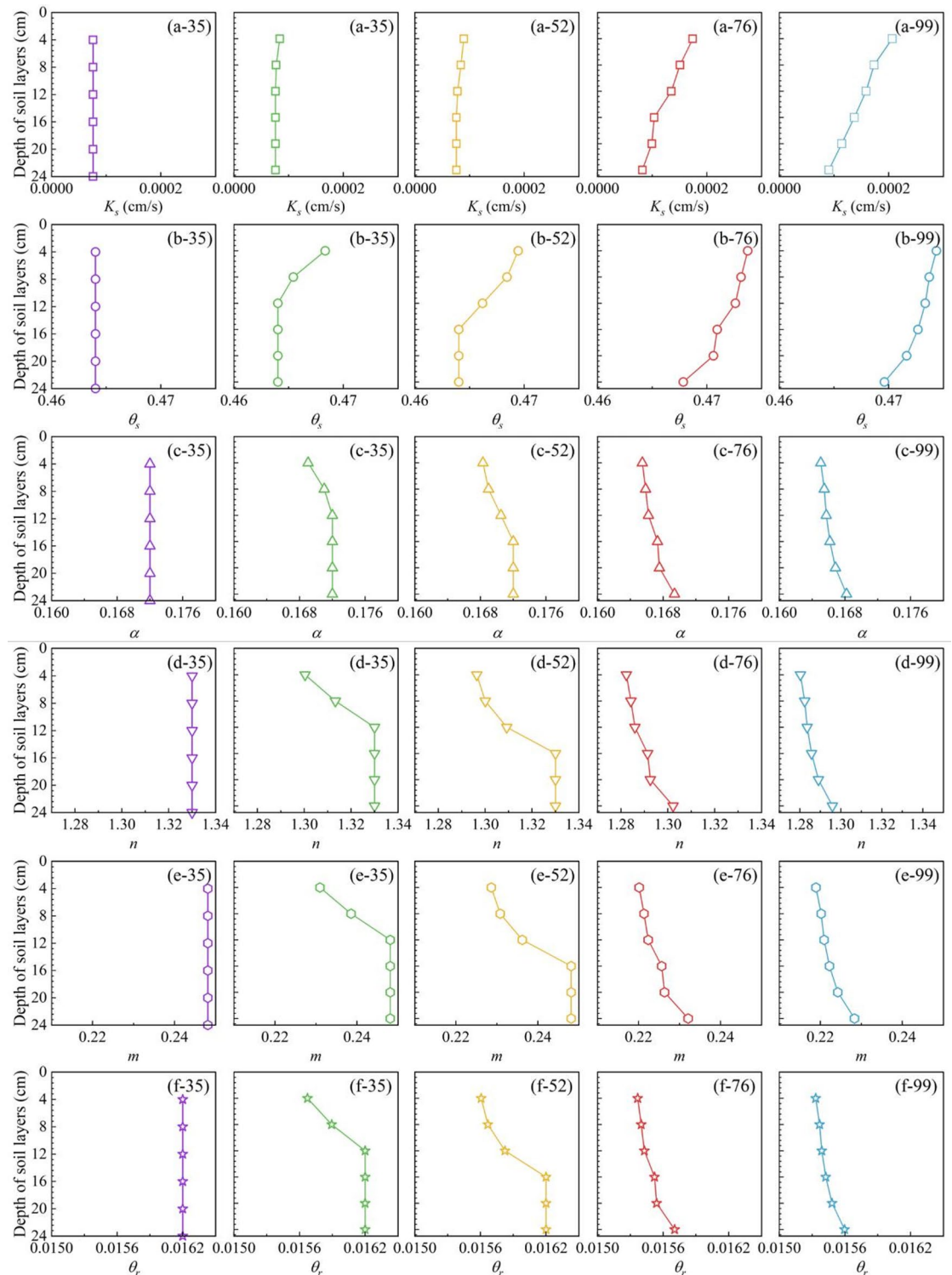


Fig. 13. The evolution law of unsaturated infiltration parameters of root-soil composite in different strata. (a): K_s , (b): θ_s , (c): α , (d): n , (e): m , (f): θ_r .

0–4 cm, K_s was approximately 1.74×10^{-4} , representing an increase of 128.96% compared with plain loess; within the 4–8 cm layer, the increase was approximately 99.38%; and within the 20–24 cm layer, the increase was the smallest, at approximately 7.8%. After the growth stage exceeded 52 d, *F. arundinacea* roots developed rapidly and formed a complex root network, and the RB at each soil layer increased significantly, thereby markedly improving the unsaturated infiltration parameters of the shallow root-soil composite. Therefore, under the growth stages of 76 and 99 d, the soil column infiltration tests showed a significant increase in wetting-front advance rate and significant decreases in total infiltration duration and sensor response time.

Temporal response analysis of water infiltration

In this study, the hydraulic parameters of bare loess and root–soil composites at different growth stages were obtained using Eqs. (7–11) in HYDRUS. Based on these parameters, numerical simulations were conducted to determine the wetting front progression and the temporal changes in volumetric water content during infiltration.

Figure 14 presents a comparison between the soil column infiltration test results and the numerical simulation results for the root–soil composites. Under different growth stage T conditions, the simulated results agreed well with the measured data. Taking the root–soil composite specimen at $T=99\text{d}$ as an example (Fig. 14(a)), the simulated wetting front reached 18.46 cm at an infiltration time of 60 min, differing from the measured value (17.75 cm) by only 0.11 cm. When the infiltration time increased to 210 min, the simulated wetting front reached 32.53 cm, with a difference of 0.52 cm from the measured value (33.05 cm). Further comparison of the total

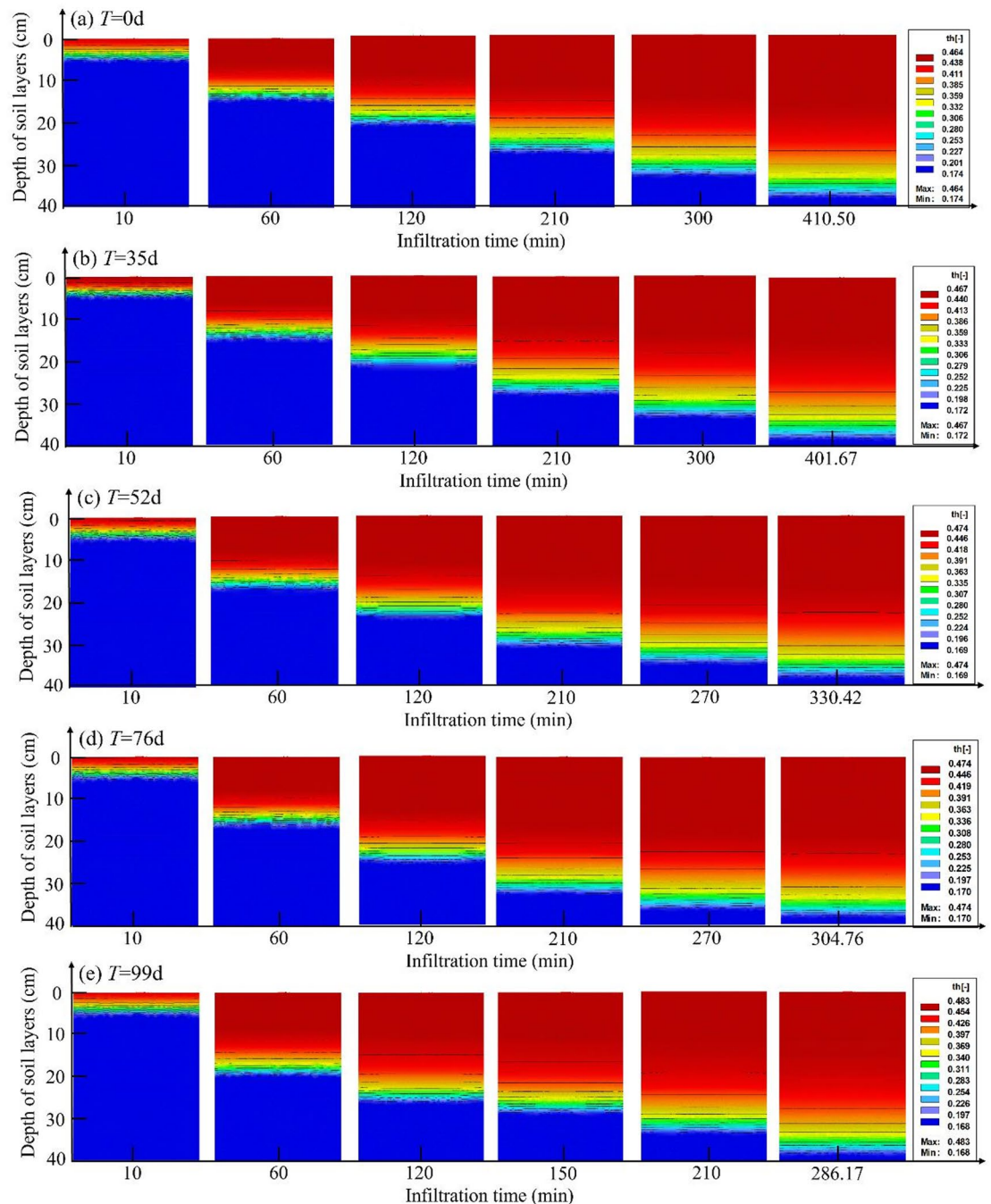


Fig. 14. The calculation results of wetting front of root–soil composite. (a): $T=0\text{d}$; (b): $T=35\text{d}$; (c): $T=52\text{d}$; (d): $T=76\text{d}$; (e): $T=99\text{d}$.

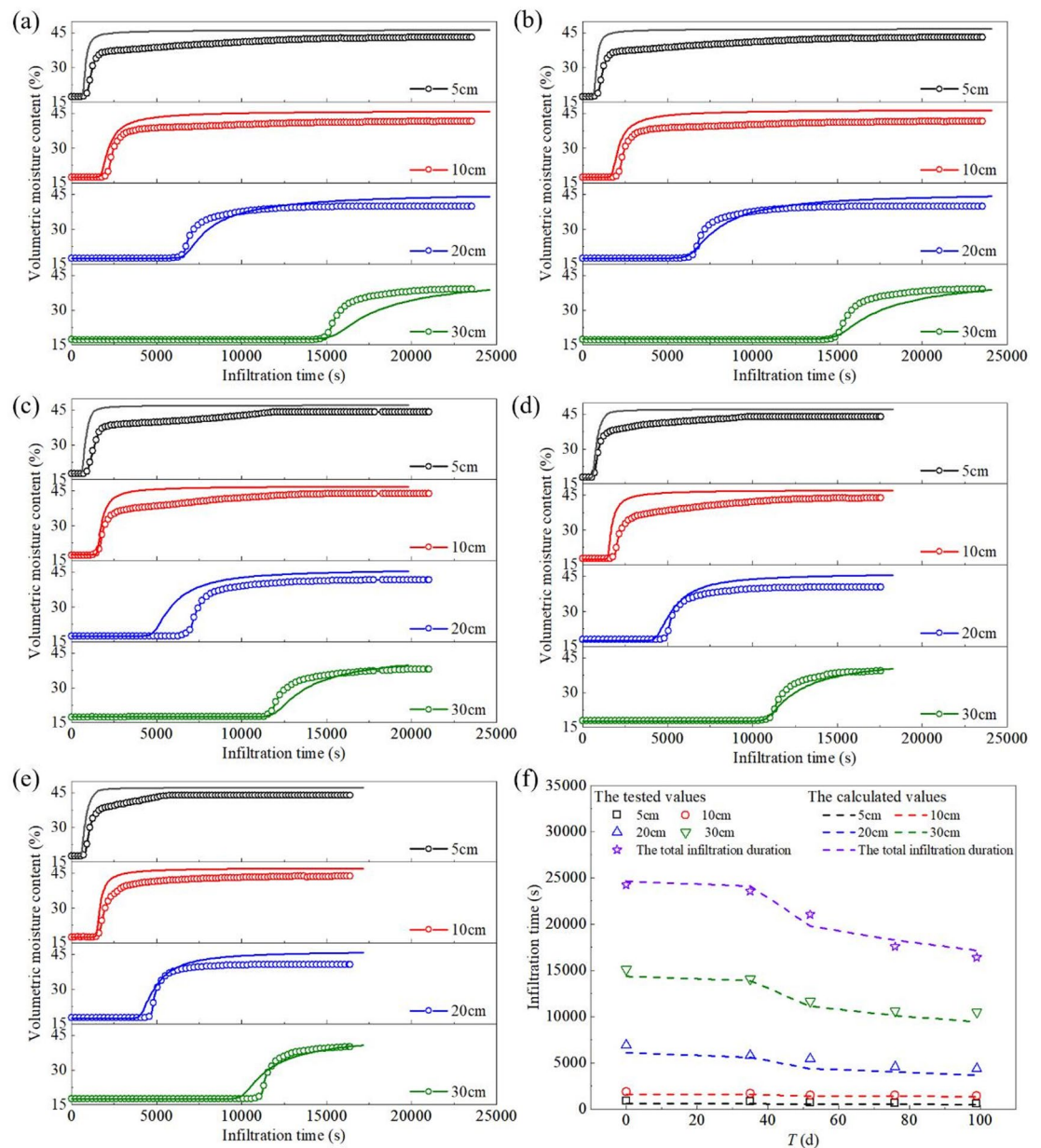


Fig. 15. Variation characteristics of volumetric water content in soil under different T . (a): $T = 0$ d; (b): $T = 35$ d; (c): $T = 52$ d; (d): $T = 76$ d; (e): $T = 99$ d; (f): S_I and infiltration time.

infiltration duration obtained from the numerical simulation, with the experimental results also showed great agreement. Both the simulated and measured total infiltration durations gradually decreased with increasing T .

Figure 15 presents a comparison between the simulated and measured time-course curves of volumetric water content under different T conditions. An overall comparison between the numerical and experimental results shows that the simulated volumetric water contents of the root–soil composites at depths of 5, 10, 20, and 30 cm were all very close to the measured values under different growth stages. The numerical results also showed a consistent trend: volumetric water content increased rapidly with infiltration time at the initial stage, then gradually approached stability, and finally remained at a nearly constant level.

Meanwhile, the simulated results at depths of 5, 10, and 30 cm showed great agreement with the measured results (Fig. 15), with differences of approximately 80–320 s. In contrast, the discrepancy between the simulated and measured values was relatively larger at a depth of 20 cm, with a difference of about 800 s. As the T increased, the root-influenced depth gradually expanded, and the RB at each soil layer also increased progressively. Accordingly, the numerical results also showed a decreasing trend in the volumetric water content response time, S_I , at all soil depths. When $T = 76$ d (Fig. 15(d)), the S_I values at the four observation points were 515.02, 1388.30, 4038.00, and 10,128.00 s, respectively. Compared with the measured values, the agreement was high at depths of 5 and 10 cm, with differences of 90.98 and 109.66 s, respectively, whereas the agreement was lower at depths of 20 and 30 cm, with differences of 517.80 and 499.80 s, respectively. When $T = 99$ d (Fig. 15(e)),

the S_I values at all four observation points decreased further and still agreed well with the measured results; however, the agreement at 30 cm was relatively lower, with a difference of approximately 1086.13 s. These minor discrepancies are mainly due to soil heterogeneity, sensor limitations, and model simplifications, while the numerical simulations still reproduce the infiltration process with high accuracy, confirming the reliability of the model.

Overall, the evolution relationships of the unsaturated hydraulic parameters proposed in this study, which incorporate the temporal effects of root growth, can be used to simulate water infiltration in root–soil composites under different growth-stage conditions and to accurately predict the temporal evolution of volumetric water content. The predicted results show great agreement with the experimental data and effectively characterize water transport behavior under dynamic infiltration conditions.

Conclusion

This study combined root distribution investigation, one-dimensional soil column infiltration experiments, and unsaturated hydraulic parameter tests to systematically analyzed the effects of herbaceous root development at different growth stages on the infiltration process and hydraulic parameter evolution of loess. The main conclusions are as follows:

- (i) An exponential function was used to quantitatively characterize the influence of vegetation growth stage on root distribution patterns. With increasing growth stage, the root distribution range first expanded and then gradually stabilized. When the growth stage exceeded =52d, the root distribution range became basically stable within the 0–24 cm soil layer. Meanwhile, the RB at different soil depths also increased rapidly at first and then gradually stabilized with increasing growth stage.
- (ii) Vegetation growth significantly affected water infiltration in loess by expanding the root distribution range and increasing RB . As the growth stage increased, the wetting-front advance rate gradually increased, whereas the total infiltration duration decreased markedly. Compared with plain loess, the total infiltration duration of the soil column under mature root conditions was reduced by up to 32.41%, indicating that root development can effectively enhance water transport capacity in loess.
- (iii) The Van Genuchten model parameters of the root–soil composite showed systematic changes with RB . Specifically, the saturated water content gradually increased and then approached a stable value, whereas the parameters α , n , m , and the residual water content generally decreased. These results indicate that root growth alters the pore structure and pore connectivity of the soil, thereby affecting the water-retention characteristics and hydraulic conductivity of loess.
- (iv) Based on the identified variation patterns of unsaturated hydraulic parameters with growth stage T , these parameter relationships were incorporated into the Richards equation to numerically simulate unsaturated infiltration under different growth-stage conditions. The HYDRUS simulation results agreed well with the soil column test results in terms of wetting-front advance and the temporal evolution of water content, indicating that the established parameter-evolution relationships can effectively reflect the dynamic influence of herbaceous root development on the infiltration process in loess.

Overall, herbaceous root growth significantly alters the infiltration behavior and hydraulic properties of loess by regulating soil pore structure and promoting preferential flow formation, and this effect shows clear growth-stage dependence. The quantitative relationships established in this study between root distribution characteristics and hydraulic parameters can provide parameter support for research on vegetation–soil–water interactions, soil water simulation, and ecohydrological assessment under vegetation restoration on the Loess Plateau.

Data availability

All data supporting the findings of this study are available within the paper and its Supplementary Information.

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Author contributions

X.C. W. and S.J. L.: collected data, interpreted the results and drafted the manuscript. P. W. and C.Y. T.: conceived the study and sequenced.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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